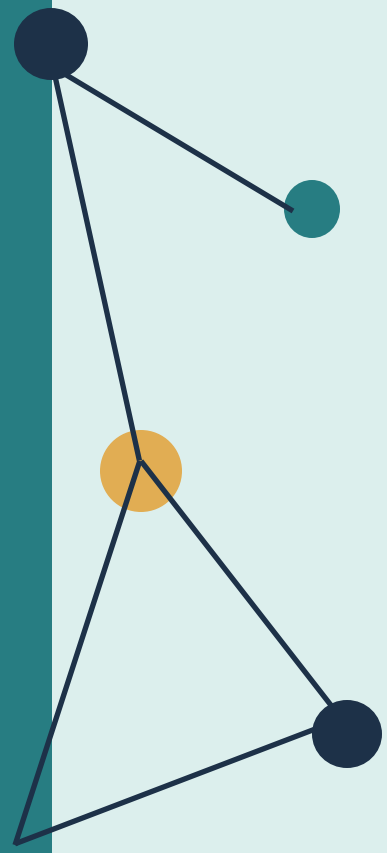
XITM6329 | PROJECT PRESENTATION

Central Karoo Interconnected District Network

A proposed municipal-owned network for hospitals, municipal offices and rural households.

Budget ceiling: R2,000,000



Prepared by: Lesedi Chewe, Owami Kanyuchi, Simam'kele Mkhonza, Neo Ngobeni, Karabelo Malinga
Rosebank International

Based on Task 1: Overview Document and Task 2: Project Charter

1. The situation in the Central Karoo

The district needs a more reliable way to connect public services and communities.

3,000 homes

The aim is to extend internet and intranet access to rural households.

3 district hospitals

Hospitals need dependable access to patient information and communication.

Municipal offices

28 offices and around 200 employees use ERP and office applications.

Where?

Western Cape, approximately 417 km southeast of Cape Town

Current problem

Fixed-line access is limited and mobile/5G connections can drop out or suffer interference.

Project idea

Build a hybrid wireless backbone that the district can manage, while still using an internet provider for upstream access.

Source: Task 1 overview and Task 2 project charter.

2. Main problems the project needs to solve

Unreliable connectivity

Unreliable 5G rollout plagued by heavy noise disconnection.

Security and supplier access

External suppliers should only reach the systems they need. Hospital and municipal data must be protected.

Power and weather

Load shedding, heat, dust and lightning can affect towers, radios and the NOC.

Limited coverage and growth

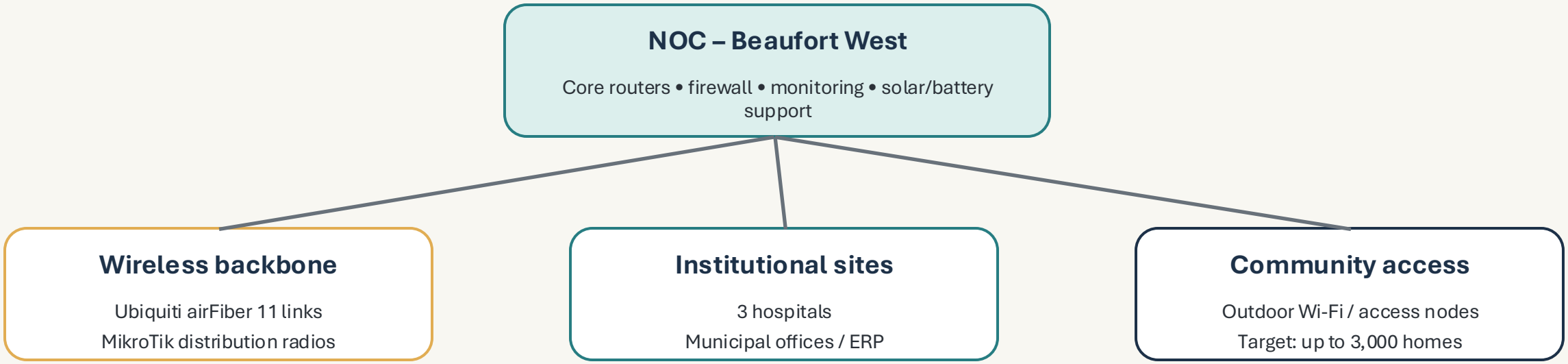
Connecting more homes may cause congestion unless capacity and bandwidth are planned properly.

The solution needs to be reliable, secure, affordable to maintain and able to expand over time.

Source: Task 1, section 6 (Potential Risks), and Task 2 technical report.

3. Proposed technical solution

A solar-powered hybrid wireless network, managed from the NOC in Beaufort West.



Network controls

VLAN 10: management | VLAN 20: municipal and healthcare | VLAN 30: public Wi-Fi
Firewall rules, VPN, FreeRADIUS access control and Zabbix monitoring

4. The Preliminary IT Solution

Component	*Current State*	Proposed Upgrade
Network Hub	*Basic Network Operating Center (NOC)*	Upgraded centralized routing hub with dynamic failover.
Topology	*Simple star configuration (two access points)*	Resilient wireless mesh and point-to-multipoint architecture.
Power	*24 solar panels & 70 deep-cycle batteries*	Upgraded off-grid integration to sustain massive scaling.

5. Project Charter and SMART Objectives

3 Months

Upgrade the NOC's routing and bandwidth capacity to handle enterprise-level traffic.

Zero

Transition the three district hospitals from the local ISP to the dedicated network with zero data loss within 4 months.

3,000

Deploy wireless bridging to connect three thousand residential homes over a 12-month phased rollout.

99.9%

Maintain 99.9% uptime leveraging the solar and battery power redundancy systems.

6. Implementation plan and deliverables

The work should happen in stages so that core services are tested before the wider rollout.

Month 1

Kick-off + design

Agree on requirements, network layout, IP plan and responsibilities.

Months 2–3

NOC upgrade

Assess equipment, procure hardware and install core network systems.

Months 4–5

Hospital/admin link

Plan data migration and connect critical public-sector sites.

Months 6–8

Home rollout – Phase A

Begin community deployment; Task 2 target: first 1,000 homes by Month 8.

Months 9–12

Home rollout – Phase B

Continue deployment towards the 3,000-home target.

Month 13*

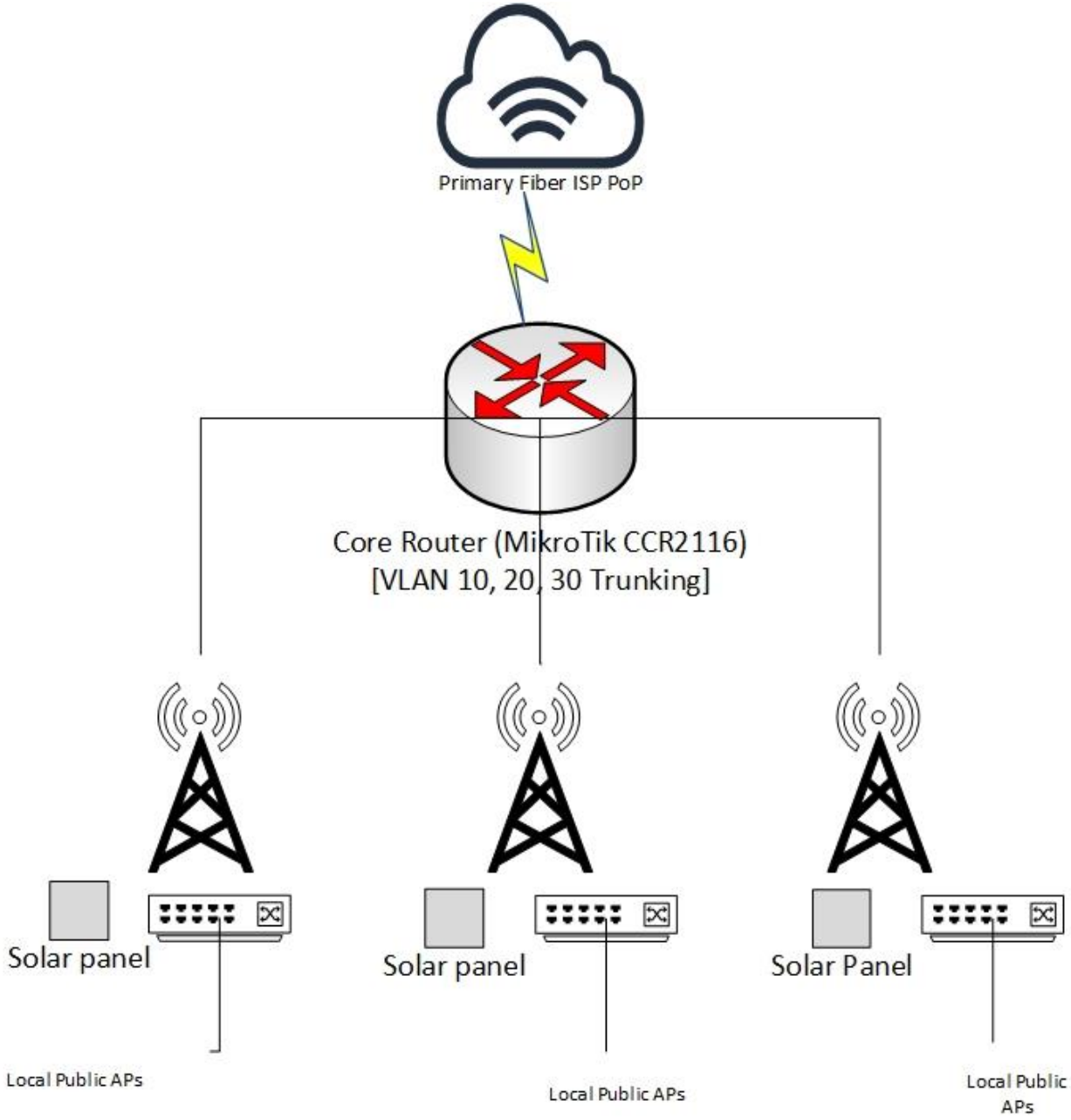
Testing + handover

Train local technicians, test the network and hand over documentation.

***Task 2 states the project goal is 12 months, but its milestone table puts final testing and handover in Month 13. Confirm the intended duration.**

Source: Task 2, Project Scope and Milestones; Key Deliverables, Timing and Resources.

7. Delivering the Architecture: The District Network Map



The central gateway is the MikroTik CCR2116. It takes the
Primary fiber feed and intelligently
Routes traffic via segmented VLAN
To dedicated local distribution towers.
This ensures critical institutional public network access.

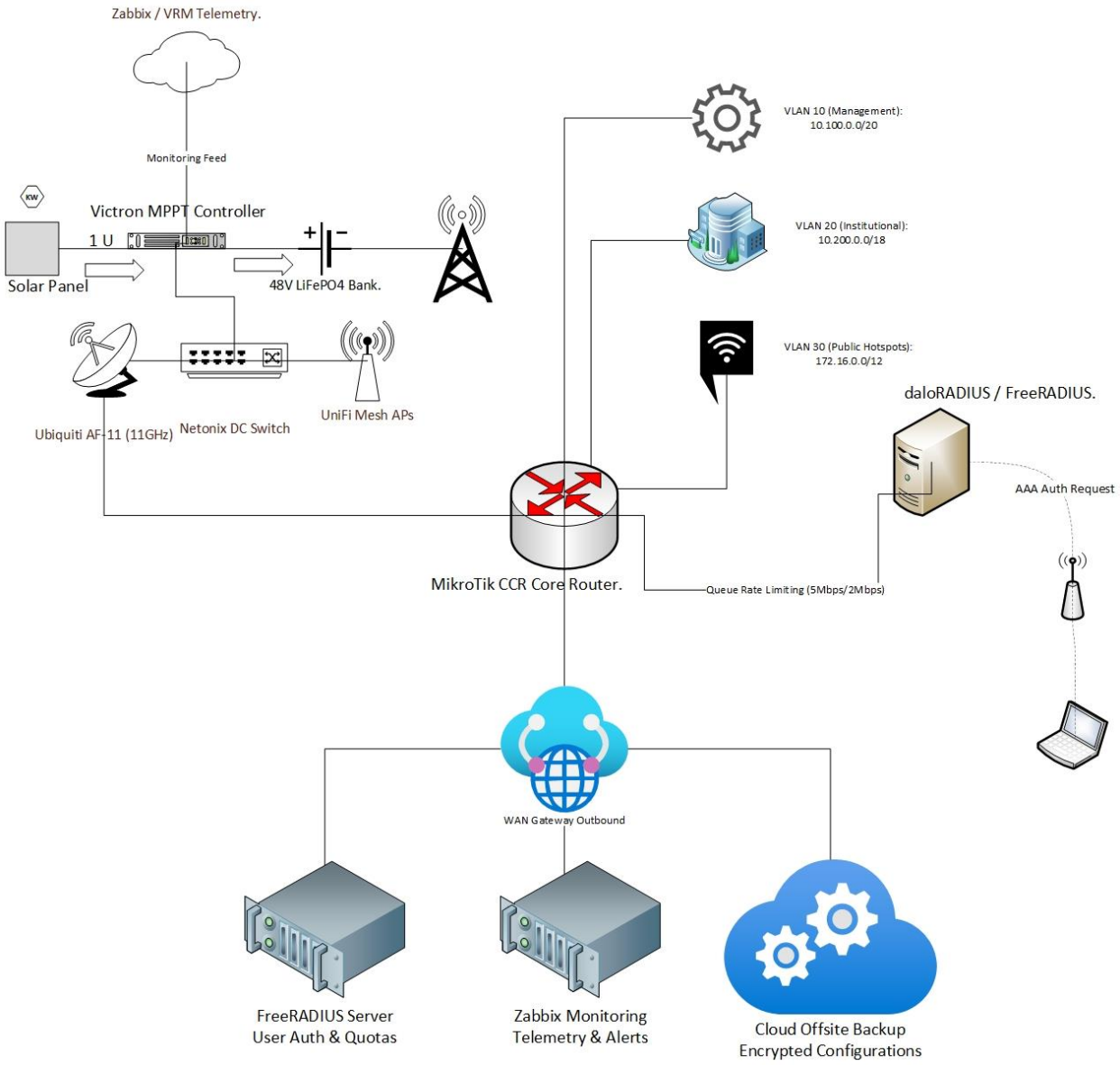
8. Delivering Security: Resolving the Trust Deficit

VLAN 10 (Management)	*For IT Admins.*	Strictly isolated. Requires encrypted SSH/VPN tunnels. No external supplier access.
VLAN 20 (Institutional)	*For Municipal Offices & Healthcare.*	Prioritized Quality of Service. Protected with enterprise-grade WPA3.
VLAN 30 (Public)	*For 3,000 Residential Homes.*	Rate-limited (5Mbps down/2Mbps up). Client isolation enforced to prevent device-to-device hacking.

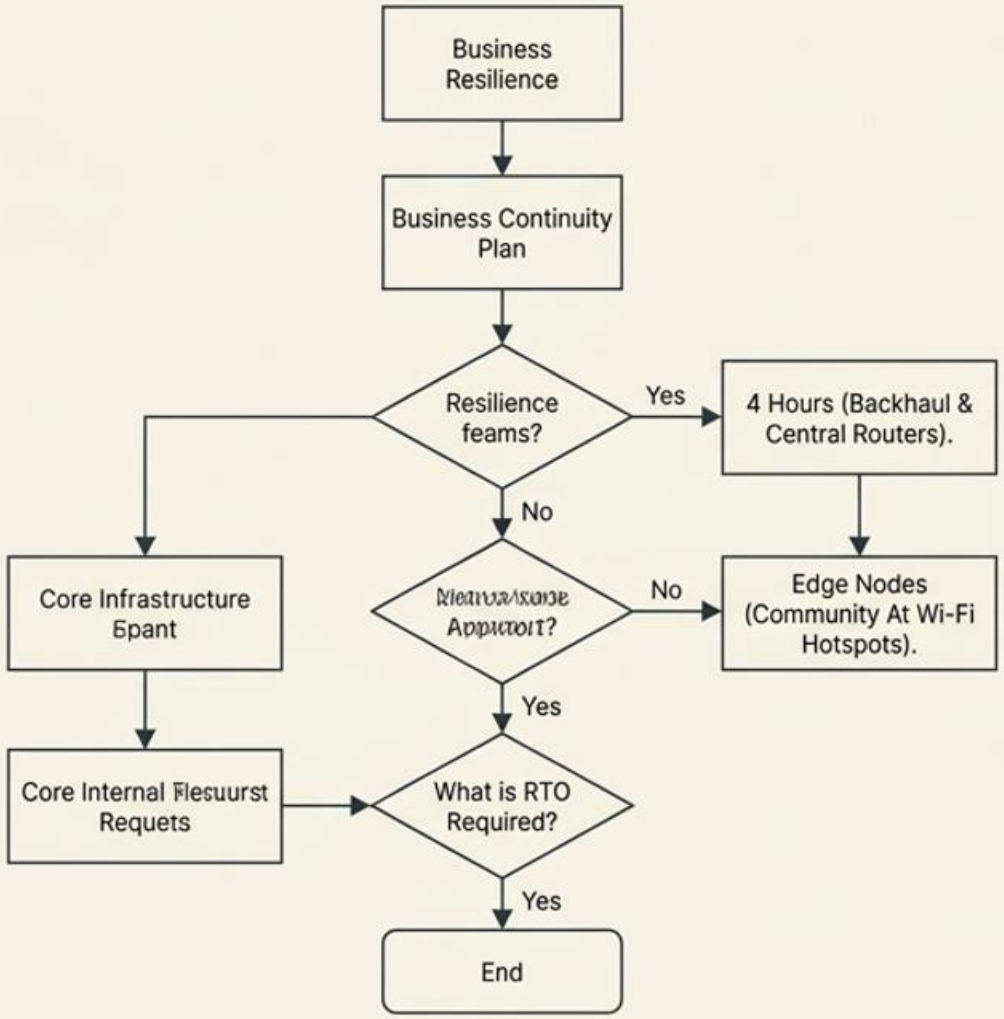
9. Delivering Security: Resolving the Trust Deficit

Figure Description: This diagram illustrates the high-level network topology for the Central Karoo District Community Network. It integrates a solar-powered off-grid backhaul and mesh Wi-Fi system, managed by a central MikroTik core router across three segmented VLANs (Management, Institutional, and Public Hotspots). The architecture incorporates AAA authentication with dynamic bandwidth rate limiting, continuous Zabbix telemetry monitoring, and secure cloud offsite configuration backups to ensure reliable community connectivity.

CENTRAL KAROO DISTRICT COMMUNITY NETWORK DIAGRAM



10. Delivering Resilience: Business Continuity Plan



Target Capability:

72-hour battery backup operation during total grid collapse.

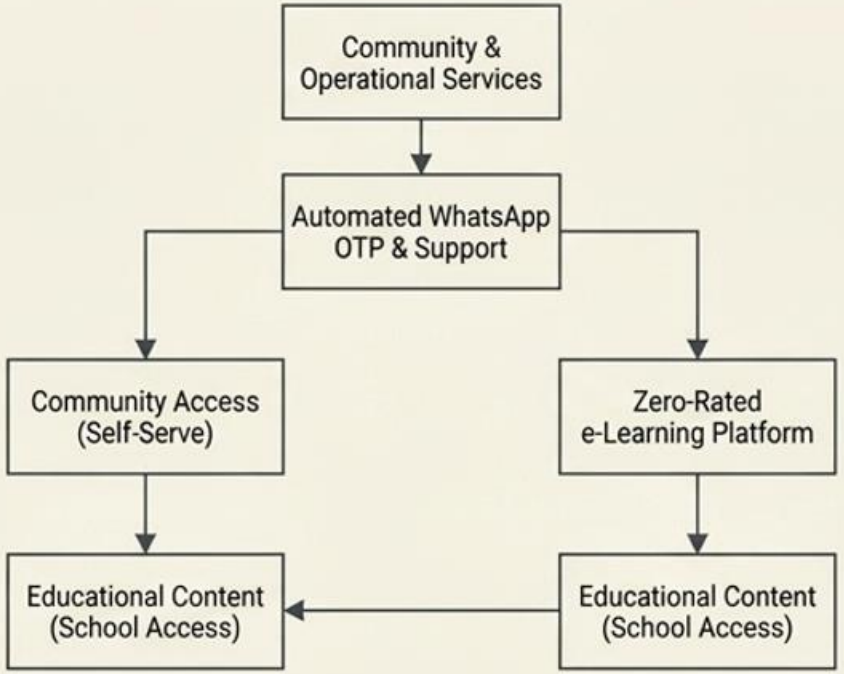
Core Infrastructure RTO:

4 Hours (Backhaul & Central Routers).

Edge Node RTO:

12 Hours (Community Access Wi-Fi Hotspots).

11. Delivering Resilience: Business Continuity Plan



By deploying automated WhatsApp OTP systems, zero-rated e-learning access, and automated ticketing, the network becomes a self-serve digital hub for the community.

12. Delivering Resilience: Business Continuity Plan

Tier 1 Fiber (Openserve/Liquid)

Securing a 1 Gbps dedicated feed guarantees 99.95% uptime without requiring the municipality to maintain physical fiber lines.

Cloud Storage (AWS)

Moving data to Cape Town data centers protects against local extreme weather and physical hardware theft.

Cybersecurity (Cybersoft)

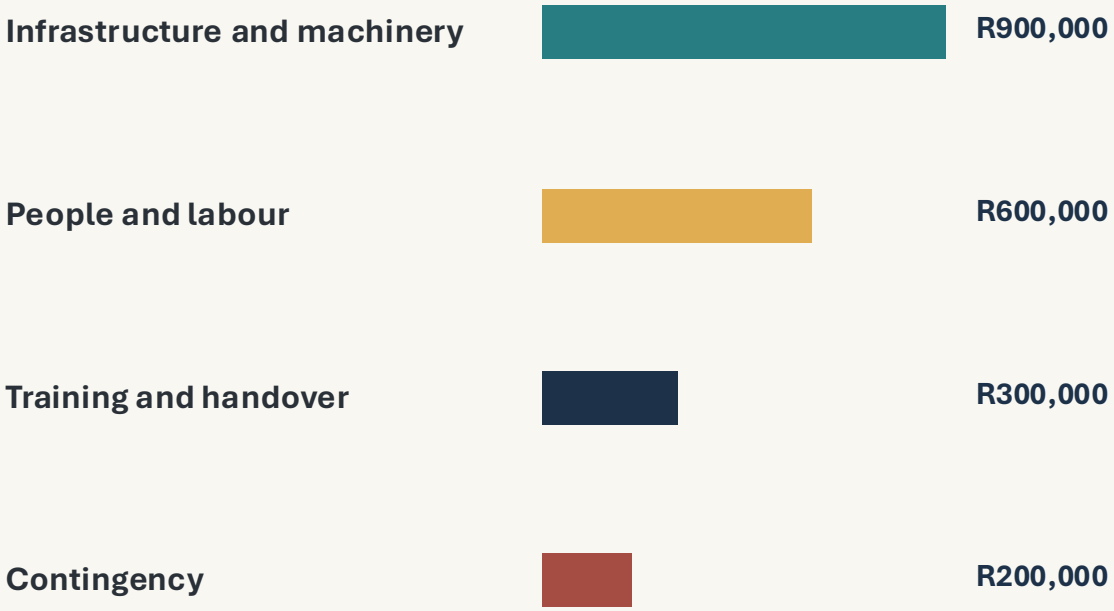
Provides independent POPIA compliance monitoring and 24/7 logging to ensure external vendors act transparently.

Specialist Rigging

Certified contractors ensure high-tower structural integrity, lightning protection, and safety compliance.

13. Budget and expected benefits

Proposed budget (ceiling: R2 million)



45% | 30% | 15% | 10%

What the project should improve

- More reliable access to hospital and municipal systems
- Less dependence on a single provider for internal connectivity
- Better control of supplier access and sensitive information
- More options for residents to access online services, learning and agricultural information
- Local staff trained to monitor and maintain the network

Uptime and full household coverage should be treated as targets that need testing and capacity planning—not automatic guarantees.

Source: Task 1 cost estimates and Task 2 budget and cost-benefit analysis.

Conclusion: Handover and Sustainability

Track 1: NOC Operations Training.

Equipping local IT staff with skills to monitor Zabbix telemetry, troubleshoot wireless links, and maintain the solar arrays.

Track 2: VIP Application Support.

Providing white-glove desktop support for the 200 government officials adjusting to the new ERP and Microsoft Office networking environment.

Establishing not just a computer network, but a self-sustaining digital foundation for the future of the Central Karoo.

Thank you